

YAVUZ BAHADIR INCI

Mechanical Engineering Student

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ABOUT

Mechanical Engineering student at Gebze Technical University ,My background in rowing and cycling sparked a deep interest in aerodynamics and fluid mechanics, driving my passion for aerospace design and innovation

EDUCATION

SEPT 2022-Ongoing

BSc Gebze Technical University, Mechanical Engineering

MAINTENANCE INTERN | ÇOLAKOĞLU METALURJİ A.Ş.

JULY2026-AUGUST2026

Supported maintenance operations for industrial mechanical equipment, including preventive maintenance, fault diagnosis, and equipment reliability improvement.
Observed maintenance practices for pumps, conveyors, gearboxes, motors, and hydraulic/pneumatic systems in a heavy industry environment.

MECHANICAL DESIGN ENGINEER | DGA DRONE

JULY2025-JANUARY2026

Led and contributed to the design, manufacturing, and development of advanced UAV projects, including Hybrid **Firefighting, Agricultural Spraying, Facade Cleaning**, and **Defense** Drones. Participated in mechanical design, CAD modeling, structural analysis, and prototype manufacturing using SolidWorks and ANSYS.
Contributed to **R&D** by developing a **glass-breaking mechanism, optimizing spray nozzle geometry**, and designing a **modular robotic arm** for industrial and defense applications.

FREELANCE MECHANICAL & SIMULATION ENGINEER

2023-Ongoing

- Delivered CFD, FEA and mechanical design services using SolidWorks, Fusion 360 and ANSYS Structural.
- Completed multidisciplinary simulation projects for engineering students and professionals.
- Developed aerodynamic CFD models for UAV platforms and atmospheric flow analyses.
- Executed architectural wind distribution studies for site planning using CFD.
- Performed structural FEA for mechanical and architectural design tasks.

Designed robotic arms, excavator mechanisms and marine vessel components in Fusion 360.

TECHNICIAN | İNCİ ELEKTRİK

2016-2022

Developed strong hands-on expertise in power tools and workshop operations, gaining a technician-oriented engineering perspective through direct involvement in installation, assembly, and maintenance tasks alongside professional electricians.

PROJECTS

RESEARCH PROJECT | Electronically Loaded Small-Scale Wind Turbine

- Developing a small-scale horizontal-axis wind turbine to investigate the effects of variable electrical loading on rotor performance and wake characteristics.
- Designing a BLDC generator system with programmable electronic load control to regulate rotor speed and tip-speed ratio (TSR).
- Integrating encoder-based RPM measurement and synchronized voltage–current data acquisition.
- Planning wind-tunnel tests and hot-wire anemometry measurements to evaluate power generation, velocity deficit, and wake turbulence.
- Preparing the project proposal within the scope of the TÜBİTAK 1002-A Rapid Support Program.

CONTROL SYSTEM DESIGN PROJECT | Dual-Rotor T-Beam Balancing System

- Designed and developed a dual-rotor balancing system for attitude stabilization using Arduino-based control.
- Developed the mathematical model and state-space representation of the system.
- Performed controller design and stability analysis using MATLAB/Simulink.
- Integrated sensors, ESCs, and BLDC motors to validate the control system on a physical prototype.
- Contributed to mechanical design, experimental validation, and technical documentation.

MECHANICAL DESIGN PROJECT | FEMOR Rehabilitation System

- Designed a modular lower-limb rehabilitation system with adjustable anthropometric dimensions.
- Performed FEA validation for 70–250 kg load cases using ANSYS Workbench.
- Optimized the mechanical design using telescopic linkages.
- Prepared and presented the complete technical report and FEA analysis.

TECHNICAL SKILLS

- FUSION 360
 - SOLIDWORKS
- ANSYS
 - MATLAB

LANGUAGE

- TURKISH | NATIVE
- ENGLISH | B2

ACTIVITIES

- GTU Rowing Crew**, Team Captain

SEPT 2023-Ongoing

Led the university's rowing team in training and national and international competitions.
Organized club events and served as president of the Rowing Club.
- GETAK Search and Rescue**, Volunteer Member

DEC 2023-2024

Participated in search and rescue training programs and emergency preparedness operations.
Collaborated with team members during simulation exercises and university-level rescue drills.
- Galatasaray Rowing Team**, Former Athlete (2 years)

Competed in national and international rowing championships, representing Galatasaray.
- Sakarya BBPRO Team**, Former Athlete (+3 years)

Competed in national and international cycling races, representing Sakarya BB Pro Team.

ORGANIZATION

- GTU Sports Summit**, Organizer

MAY 2025

Coordinated and managed the GTU Sports Summit, a multidisciplinary event bringing together student-athletes, coaches, and professionals to discuss the intersection of sports, science, and leadership.

 - Oversaw event planning, speaker invitations, and social media outreach

HOBBIES

I actively create educational video content on Instagram and YouTube, focusing on aircraft, aerospace systems, engineering concepts, coaching insights, and popular science topics. I take pride in making complex ideas more accessible to a broader audience. My background as a former professional athlete in rowing (Galatasaray Rowing Team) and cycling (Sakarya BB Pro Team) sparked a strong passion for aerodynamics and flight mechanics, shaping my long-term interest in aerospace engineering. In addition, I am deeply interested in scientific research and regularly analyze new studies to stay informed about the latest advancements in science and technology.